

An Asymmetric Formula for Interval Consonance and its Relation to Harmonic Coincidence

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Abstract

Euler’s *Gradus Suavitatis* (1739) assigns a dissonance value to a musical interval p/q by the formula $G(p/q) = 1 + \Omega^*(p) + \Omega^*(q)$, where $\Omega^*(n) = \sum_i e_i(p_i - 1)$ sums the weighted prime exponents of n . On a standard set of 13 just intervals, the simpler asymmetric formula $f(p/q) = p + \Omega^*(q)$ attains a slightly higher rank correlation with published human consonance ratings ($\rho = 0.989$ vs. 0.979). We also show that, under a model in which harmonics are integer-indexed and counted uniformly up to a fixed truncation level, Gradus is equivalent to a weighted harmonic coincidence count with weights $w(n) = \Omega^*(n)$, connecting it to Galileo’s earlier pulse-coincidence model (1638). The formula naturally generates a coprime integer triangle $T(n, k) = n + \Omega^*(k)$, whose rightmost diagonal gives the two-stage dissonance of the superparticular (consecutive-harmonic) intervals. The formula f admits a simple two-stage interpretation in terms of harmonic context and partial recognition, which we offer as a conjecture.

1 Background

Euler’s *Gradus Suavitatis* [6], catalogued by the author in OEIS A275314, assigns to each musical interval a positive integer measuring its arithmetic dissonance. For a ratio p/q in lowest terms, the characteristic number is $n = p \cdot q$, and:

$$G(p/q) = 1 + \Omega^*(p \cdot q)$$

where Ω^* is the completely additive function $\Omega^*(n) = \sum_i e_i(p_i - 1)$. Since Ω^* is completely additive, $G(p/q) = 1 + \Omega^*(p) + \Omega^*(q)$, treating numerator and denominator symmetrically. The weights $(p_i - 1)$ for prime p_i encode Euler’s aesthetic judgment that large primes represent greater “difficulty.”

Euler’s formula correctly ranks the 13 standard intervals of Western music against human consonance ratings [1] with Spearman $\rho = 0.979$. We note it is exactly tied on the major third/major sixth pair and on a three-way tie at Gradus 8, while these are distinct values in the human consonance rank.

The 13 intervals are the standard dyads of the Western chromatic and diatonic scales shown in Table 2. The human ratings follow Krumhansl [1], which reports data from musically trained Western listeners and assumes octave equivalence. With only 13 data points, rank correlations should be read as illustrative comparisons rather than statistically decisive results; the ordering is well-replicated [2, 3] and broadly consistent with cross-cultural data [4, 5], though precise values depend on musical training and cultural exposure.

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Organisation. Section 2 shows that Gradus admits an exact reinterpretation as a weighted harmonic coincidence count. Section 3 introduces $\max(p, q)$ and its four classical interpretations. Section 4 presents the asymmetric formula $f(p/q) = p + \Omega^*(q)$. Section 5 compares all three formulae in detail. Section 6 examines the one tie that neither f nor $\max$ can resolve. Section 7 conjectures a two-stage perceptual interpretation of f . Section 8 surveys the consonance landscape, established scales, and equal temperament systems, with the caveat that a full treatment would need to consider the timbre and tuning of instruments. Section 9 develops a partial-beating tolerance model that connects the arithmetic formula to acoustic roughness and register dependence, and reveals a duality between the perceptual cost of an interval and its sensitivity to mistuning. Section 10 summarises.

2 Euler’s Gradus as Harmonic Coincidence

Throughout this paper the harmonic series is treated as a discrete, integer-indexed model: harmonics are indexed by positive integers m , counted uniformly, and spectral envelopes, inharmonicity, and equal temperament deviations are set aside. The derivation below is mathematically exact within this idealisation; its applicability to real musical sounds — which have finite and instrument-dependent spectra, slight inharmonic stretch, and other deviations from the ideal — is the empirical question that the correlation data address.

For two notes at frequency ratio $p : q$ (coprime), the m -th harmonic coincidence occurs at absolute frequency $mp \cdot f_0$, which is the (mp) -th harmonic of the lower note and the (mq) -th harmonic of the upper note. The following derivation assumes a fixed truncation level M , the same for all intervals, and that each coincidence event contributes equally to the score. In practice M is determined by the spectral content of the sound and the frequency range of hearing, and may vary with interval; the identity below is therefore a model-dependent result, not a universal physical law.

Assigning weight $w(n) = \Omega^*(n)$ to the n -th harmonic of each note, the weighted contributions to the lower and upper note respectively, summed over M coincidences, are:

$$\begin{aligned} \text{Lower: } \sum_{m=1}^M \Omega^*(mp) &= \Omega^*(p) \cdot M + C(M) \\ \text{Upper: } \sum_{m=1}^M \Omega^*(mq) &= \Omega^*(q) \cdot M + C(M) \end{aligned}$$

where $C(M) = \sum_{m=1}^M \Omega^*(m)$ is independent of both p and q . Both identities follow from the complete additivity of Ω^* : $\Omega^*(mn) = \Omega^*(m) + \Omega^*(n)$ for all positive integers m, n . This is an arithmetic identity, not a physical one: it is because Ω^* is completely additive that the multiplicative structure of prime factorisation and the additive structure of harmonic series indexing happen to be compatible. This arithmetic coincidence is what makes the model tractable; it is a modelling bridge rather than a physical necessity. Adding the two sums gives the symmetric score:

$$\text{Score} = M \cdot (\Omega^*(p) + \Omega^*(q)) + 2C(M) = M \cdot (G(p/q) - 1) + 2C(M)$$

Gradus therefore admits, within this model, an exact reinterpretation as a weighted harmonic coincidence count, with harmonics weighted by their own prime complexity. This confirms the physical intuition behind Euler’s formula but reveals its self-referential structure: recovering Gradus from coincidences requires weights that are themselves Gradus-like. We emphasise that this is a reinterpretation of Gradus rather than Euler’s own motivation, which was aesthetic and number-theoretic.

Unweighted coincidence counting (flat weights) achieves $\rho = 0.791$ across all coprime ratios with $p, q \leq 16$. The weight function $\log n$ achieves $\rho = 0.795$ without any prime structure, establishing a ceiling for physics-based models that do not invoke prime factorisation.

3 Galileo’s Model and $\max(p, q)$

Galileo [7] described consonance in terms of pulse coincidences: two strings at ratio $p : q$ produce coinciding pulses every p vibrations of the lower string. The “sweetness” of the interval grows with the frequency of coincidences, i.e. inversely with p . The metric $\max(p, q) = p$ is therefore:

1. **Galileo’s pulse model** — vibrations between coincidences;
2. **Harmonic series position** — the upper note is partial p of the implied fundamental [8];
3. **Farey index** — the smallest n such that q/p appears in the Farey sequence F_n (Farey 1816; proved by Cauchy);
4. **First coincidence harmonic** — the position in the lower note’s series where the two notes first meet.

These four descriptions, discovered independently across four centuries, are mathematically identical. $\max(p, q)$ achieves $\rho = 0.989$ against human ratings, marginally exceeding Gradus, but cannot distinguish intervals with the same numerator (e.g. $5/4$ and $5/3$ both have $\max = 5$).

Galileo’s account predates Euler’s *Gradus Suavitatis* by a century, and is physical rather than arithmetic: it concerns the periodic coincidence of string vibrations, not the factorisation of integers. It is therefore significant that the purely arithmetic measure $\max(p, q)$, and more so the formula $f(p/q) = p + \Omega^*(q)$ of Section 4, reproduce the same ordering as Galileo’s dynamical model. This suggests that prime-arithmetic complexity and vibration period carry equivalent information about perceived consonance, and that the connection between physics and arithmetic here is deeper than coincidence.

4 A Simpler Asymmetric Formula

Throughout this paper, p/q denotes a frequency ratio in lowest terms with $p > q \geq 1$ and $\gcd(p, q) = 1$, so that $p/q > 1$ represents an interval above unison and p is always the numerator (upper note).

Euler’s formula is symmetric: $\Omega^*(p) + \Omega^*(q)$. Human ratings, however, distinguish numerator and denominator — the bass note and the upper note play different perceptual roles. We find empirically that the asymmetric formula:

$$\boxed{f(p/q) = p + \Omega^*(q)} \quad (p > q \geq 1, \gcd(p, q) = 1)$$

achieves $\rho = 0.989$ on the 13-interval dataset, matching $\max(p, q)$ and slightly exceeding Gradus. The formula differs from Gradus only in replacing $\Omega^*(p)$ with p for the numerator.

The formula naturally generates a coprime integer triangle

$$T(n, k) = n + \Omega^*(k), \quad 1 \leq k \leq n, \gcd(n, k) = 1,$$

whose rightmost diagonal $T(n, n-1)$ gives the two-stage dissonance of the superparticular (consecutive-harmonic) intervals $n/(n-1)$. Both the triangle and its diagonal are treated in detail in Section 10.

The choice of $\Omega^*(q)$ for the denominator term is motivated by its role in Euler’s Gradus and by the correspondence with human data; however, other complexity measures of q — for example $\log q$, the unweighted prime count $\Omega(q)$, or the largest prime factor — yield structurally similar but numerically distinct formulae.

The formula $f(p/q) = p + \Omega^*(q)$ decomposes as the sum of two quantities:

Component	Formula	Meaning
Upper note reach	p	Partial number of upper note in implied harmonic series
Bass context depth	$\Omega^*(q)$	Prime complexity of lower note’s position in series

Section 7 develops a conjectured perceptual story behind this decomposition.

5 Comparison of the Three Formulae

Table 2 invites a direct comparison of the three arithmetic formulae. Several observations follow.

Agreement on the extremes. All three formulae agree without qualification on the most and least consonant intervals. The unison (1), octave (2), fifth (3/2), and fourth (4/3) are assigned the lowest values by every measure, consistent with their historical priority in counterpoint and their prominence in non-Western musical systems. The tritone (45/32) receives the highest value under every formula, and the minor and major seconds cluster near the top of the dissonance scale in all three cases. The disagreements are concentrated in the middle of the table, among intervals that are also the most contested in historical music theory.

Correlation and ties. Euler’s Gradus achieves a Spearman rank correlation of $\rho = 0.979$ against the human data; f and $\max$ both achieve $\rho = 0.989$. The difference is modest, and caution is warranted given the small sample (13 intervals) and the cultural specificity of the rating data (see Section 6). More informative than the correlation coefficient is the tie structure. Gradus produces two tie groups among the 13 intervals: the major third and major sixth share $G = 7$, and the minor third, minor sixth, and major second share $G = 8$. The three-way tie at $G = 8$ is particularly awkward, as the three intervals span three consecutive human ranks (6, 8, 9). The formula f resolves this three-way tie — giving $f = 10, 12, 12$ respectively — though it introduces a two-way tie between the minor sixth and major second (both $f = 12$, human ranks 8 and 9). The metric $\max(p, q)$ shares the major third / major sixth tie with f , but adds a further tie between the major second and minor seventh (both $\max = 9$, human ranks 9 and 10). On the criterion of tie-resolution among these 13 intervals, f performs best, Gradus intermediate, and $\max$ least well.

Dynamic range and the tritone. The three formulae differ most sharply in their treatment of the tritone. Gradus gives $G(45/32) = 14$, only four units above the major seventh ($G = 10$). Both f and $\max$ assign substantially higher values ($f = 50$, $\max = 45$), placing the tritone in a category of its own and reflecting the large numerator $p = 45$ that the asymmetric formulae weight directly. Whether this wider dynamic range better represents human experience is difficult to assess from rank-correlation data alone; what can be said is that the tritone is consistently rated as by far the most dissonant interval in the human data, and both f and $\max$ capture that separation more emphatically than Gradus.

A caveat is necessary: the tritone used throughout this paper is 45/32, the 5-limit augmented fourth generated within the diatonic scale as $45/32 = (9/8) \cdot (5/4)$. This choice is conventional in 5-limit just intonation but is not the only defensible one. The septimal tritone

7/5 (582.5 cents) gives $f(7/5) = 7 + \Omega^*(5) = 11$ and $G(7/5) = 11$ — barely more dissonant than the major third under either measure. The undecimal tritone 11/8 gives $f = 14$. These alternatives sit in different prime limits and carry different musical connotations: 7/5 is the tritone of barbershop and some jazz practice; 45/32 is the tritone of classical diatonic harmony. The extreme values $f = 50$ and $\max = 45$ are therefore properties of the 5-limit diatonic tritone specifically, not of the tritone concept in general, and should be interpreted accordingly. The high values are, however, consistent with the 5-limit diatonic context in which the human rating data were collected.

Symmetry. Gradus is symmetric in p and q : $G(p/q) = G(q/p)$. This reflects Euler’s focus on the product $p \cdot q$ as the relevant arithmetic quantity, and is appropriate if the perceptual roles of the two notes are interchangeable. Both f and $\max$ are asymmetric, treating the numerator (upper note) and denominator (lower note) differently, as is the case when one note functions as a bass and the other as a melody note. The symmetry of Gradus means it cannot distinguish 9/5 from 16/9 — two minor sevenths that differ by only 1 cent but arise from different tuning systems — whereas f gives $f(9/5) = 13$ and $f(16/9) = 20$. Whether this asymmetry reflects a genuine perceptual distinction or an artefact of the formula is an open question; the conjecture of Section 7 offers one possible motivation.

Interval inversion. Two intervals whose frequency ratios multiply to 2 are *inversions* of each other within the octave: the major third (5/4) inverts to the minor sixth (8/5), the fifth (3/2) to the fourth (4/3), and so on. Under Gradus and under Tenney height, inversionally related intervals receive similar but not identical values ($G(5/4) = 7$ vs $G(8/5) = 8$; $H(5/4) = 20$ vs $H(8/5) = 40$). The formula f is more strongly asymmetric: $f(5/4) = 7$ but $f(8/5) = 12$, a gap of 5 units. This captures the perceptual asymmetry — a minor sixth in root position (bass on the lower note) is generally heard as less stable than a major third — but amplifies it considerably. Whether the magnitude of the asymmetry is empirically justified would require testing beyond the Krumhansl dataset, which does not distinguish root-position from inverted voicings.

Tenney height. The Tenney height $H(p/q) = p \cdot q$ [9], also known as the Benedetti height, is included in Table 2 for comparison. It achieves $\rho = 0.978$ against the human data, comparable to Gradus and slightly below f and $\max$. Tenney height is symmetric like Gradus, but multiplicative rather than additive: it measures the prime complexity of the product $p \cdot q$ without distinguishing numerator from denominator. Its ordering of the 13 intervals differs from human judgments most sharply on the major third / major sixth pair: $H(5/4) = 20 > H(5/3) = 15$, placing the major sixth above the major third in consonance — directly opposite to human ratings. This failure is qualitatively different from the tie produced by f and $\max$: all additive prime-weighted formulae tie 5/4 and 5/3 because $\Omega^*(4) = \Omega^*(3)$, whereas Tenney actively inverts them. The result supports the view that a multiplicative symmetric measure is not the right structure for this problem.

Simplicity and interpretability. The metric $\max(p, q)$ is strikingly simple, requires no prime factorisation, and admits four independent mathematical characterisations (Section 3). These properties make it an attractive baseline. Its weakness is precisely that simplicity: it is insensitive to whether the denominator is a power of 2 or a large prime, treating 4/3 and 7/3 identically in the denominator. Gradus captures this prime sensitivity through $\Omega^*(q)$ but applies the same weighting to the numerator, where the evidence suggests a different treatment is warranted. The formula f retains prime sensitivity in the denominator while replacing the numerator term with the raw partial number, as is motivated by the conjecture of Section 7.

Each formula thus represents a different theory of what makes an interval consonant, and the empirical data do not decisively separate them.

Sensitivity near just ratios. Figure 1 illustrates these differences directly. For two showcase intervals — the fifth ($3/2$) and the major third ($5/4$) — every rational number with denominator ≤ 32 within ± 40 cents of the just ratio is plotted against its formula value. The just ratio sits at the centre of a deep trough, surrounded by rationals with substantially higher values under all three formulae. The red markers (f) are consistently the tallest: moving roughly 20 cents from the just fifth, f reaches values of 70 or above while Gradus (G , blue) remains in the 40–55 range. The green markers ($\max$) are intermediate but track the numerator p directly and so cluster at values that increase smoothly with denominator size. Within the ± 15 cent tolerance zone (gold shading), no rival rational with denominator ≤ 32 exists for either interval, confirming that the just ratios are genuinely isolated at low complexity in their neighbourhood.

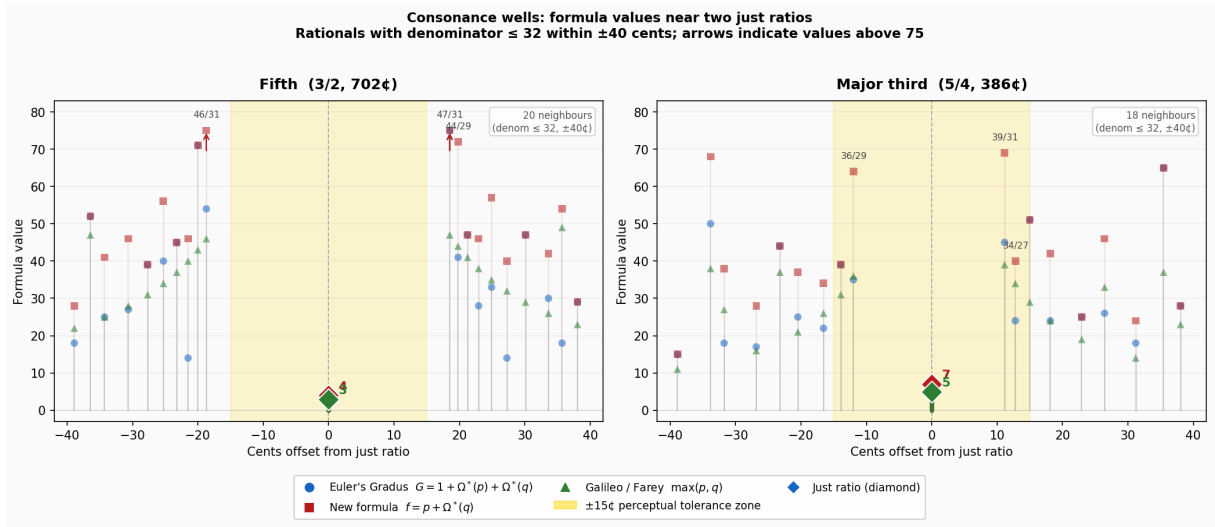


Figure 1: Formula values for all rationals with denominator ≤ 32 within ± 40 cents of the just fifth ($3/2$, left) and major third ($5/4$, right). Diamond markers show the just ratio; circles, squares, and triangles show neighbouring rationals under G , f , and $\max$ respectively. Upward arrows indicate values above 75. The gold band marks the ± 15 cent perceptual tolerance zone; no rival rational falls within it for either interval.

Choice of formula. The three formulae are more complementary than competing. For a context requiring only a rough ordering of standard intervals, any of the three suffices. Where fine distinctions matter — competing tuning systems, intervals outside the standard 13, or analysis of the consonance landscape under equal temperament — the formulae diverge in informative ways, and the choice should be guided by the theoretical question at hand. Harrison & Pearce [10] provide a comprehensive computational framework decomposing simultaneous consonance into roughness, harmonicity, and cultural familiarity components; the formula $f(p/q)$ may be understood as an arithmetic approximation to the harmonicity component of that decomposition, with the partial-beating tolerance model of Section 9 corresponding to the roughness component.

6 Limitations: the Major Third and Major Sixth Tie

No simple formula in this class distinguishes the major third ($5/4$, human rank 5) from the major sixth ($5/3$, human rank 7). Both have $p = 5$ and $\Omega^*(q) = 2$ (since $\Omega^*(4) = \Omega^*(2^2) = 2$ and $\Omega^*(3) = 2$). The distinction requires knowing that $q = 4 = 2^2$ is simpler than $q = 3$ under octave equivalence, yet the prime-weighting function cannot see this.

The Tenney height $H(p/q) = p \cdot q$ [9] — the standard complexity measure in just intonation theory — makes the situation worse: $H(5/4) = 20 > H(5/3) = 15$, so Tenney ranks the major sixth as *more consonant* than the major third, the opposite of human judgments. Gradus, f , and max all tie the pair; Tenney inverts them. This failure of Tenney height is a direct consequence of its symmetric and multiplicative structure: $H(p/q) = p \cdot q$ is dominated by whichever of p and q is larger, which for $5/3$ is $q = 3$ (small) rather than for $5/4$ where $q = 4$ (larger). The asymmetric formula f ties them for the same structural reason — $\Omega^*(3) = \Omega^*(4) = 2$ — but does not invert the order.

Lahdelma & Eerola [2] decompose consonance ratings into sensory and familiarity components; on *sensory consonance* alone, $5/4$ and $5/3$ are nearly indistinguishable across listeners, suggesting that the tie in $f(p/q)$ may accurately reflect the purely arithmetic component of the percept. A subsequent study [3] further shows that register and inversion affect consonance ratings independently of roughness and harmonicity, providing direct empirical support for the asymmetric role of the bass note assumed in Section 7. The gap observed in Krumhansl’s data is plausibly attributable to learned exposure — major thirds occur far more frequently than major sixths in tonal music — consistent with the finding of McDermott et al. [4] that consonance preferences in a culture with minimal Western music exposure are substantially reduced. The converse view — that consonance preferences reflect universal statistics of harmonic sounds in the environment, including voiced speech — is advanced by Bowling, Purves & Gill [5], who find that vocal similarity predicts interval attraction across cultures; on this account the residual consonance of ET thirds may reflect their partial alignment with vocal harmonic spectra. Resolving the tie within an arithmetic formula would therefore require modelling cultural familiarity, which is outside the scope of the present approach.

7 A Two-Stage Perceptual Interpretation

We conjecture an explanation of the formula by considering a two-stage model of how the auditory system may evaluate an interval:

Stage 1 — Bass establishes harmonic context (cost $\Omega^*(q)$). The lower note is the q -th partial of an implied fundamental F . The brain extrapolates downward to F by “dividing out” the prime factors of q . Each factor of 2 (an octave) costs 1 unit; each factor of 3 costs 2; each factor of 5 costs 4. This is exactly $\Omega^*(q)$. Intervals whose bass note is a power of 2 (e.g. the octave, $q = 1$; the major second bass, $q = 8 = 2^3$) establish the fundamental cheaply via octave equivalence. Intervals with an odd prime in q (the fourth, $q = 3$; the minor third bass, $q = 5$) require the brain to extrapolate through a genuinely new prime, incurring higher cost. The mechanism of subharmonic matching — by which the auditory system infers a missing fundamental whose upper partials best account for the heard tones — is formalised by Parncutt [11], extending Terhardt’s virtual pitch theory to chord roots; Stage 1 of our conjecture draws on this framework. In that framework the brain does not literally factorise integers: it performs pattern matching on harmonic templates or, equivalently, autocorrelation of the acoustic waveform across candidate periods. The prime decomposition in our formula acts as an *analytical proxy* for this biological process: $\Omega^*(q)$ counts, in a principled arithmetic way, the number of octave-reduction and prime-ascent steps needed to reach the implied fundamental, mirroring the complexity of the template search without claiming that the auditory system performs explicit prime factorisation.

Stage 2 — Upper note reaches into the series (cost p). Once the fundamental is established, the upper note must be recognised as its p -th partial. Higher partials are quieter (amplitude $\propto 1/p$ for harmonic sounds with a falling spectral envelope, such as bowed strings and the singing voice; the exact roll-off is instrument-dependent), more narrowly tuned, more sensitive to mistuning, and have lower pitch salience [12]. The cost of confirming the upper note as a member of the harmonic series is therefore modelled as p itself.

Conjectured total perceptual effort: $f(p/q) = p + \Omega^*(q)$.

This conjecture is consistent with:

- **Galileo** (1638): pulse coincidences as physical mechanism;
- **Rameau** (1722): the *corps sonore* (harmonic series) as the source of musical meaning;
- **Euler** (1739): prime arithmetic as the language of harmonic complexity;
- **Terhardt** (1979): virtual pitch and harmonic template matching as auditory mechanisms;
- **Huron** [13]: statistical learning of harmonic patterns as the basis of consonance judgments.

We note that the formula $p + \Omega^*(q)$ is not the unique arithmetic expression consistent with this interpretation: other choices such as $p + \log q$ or $p + \Omega(q)$ (unweighted prime count) are structurally plausible. The particular fit of $\Omega^*(q)$ — Euler’s prime-weighted function — to human data is an empirical observation rather than a theoretical necessity, and invites further experimental investigation.

8 The Consonance Landscape and Equal Temperament

The three formulae are defined on rational numbers, but musical intervals in practice are never exact. To study how each formula behaves near a just ratio we sweep the octave in 1-cent steps, find the simplest rational approximation p/q (denominator ≤ 64) within ± 15 cents of each position, and evaluate G , f , and $\max$. The result is a *consonance landscape* — a spiky function of pitch interval in which low values cluster sharply around simple fractions and rise steeply on either side.

Well depth and sharpness

Table 1 shows, for each standard just interval, the nearest rival rational within ± 50 cents and the corresponding jump in each formula. The wells around simple fractions are deep for all three functions, but $f(p/q)$ is substantially steeper: moving 2–5 cents off the fifth ($3/2$) takes f from 4 to 104 (+100), compared with Gradus rising from 4 to 58 (+54). The major third is even more extreme: 3 cents off $5/4$, f leaps from 7 to 136, while Gradus rises from 7 to 81. This sharper gradient means $f(p/q)$ suggests tighter perceptual boundaries around just intonation than Gradus does — a potentially testable difference.

Equal temperament

Twelve-tone equal temperament (12-TET) divides the octave into 12 equal semitones of 100 cents each; no interval except the unison and octave is exactly just. Table 3 in Appendix B lists all 12-TET semitones alongside their 5-limit just intonation targets and formula values. The pattern is clear: the fourth and fifth have ET errors of only 2 cents and low formula values; the thirds and sixths have errors of 14–16 cents and moderate values; the minor second and tritone have both large errors and the highest complexity values. Each ET semitone relies on

Table 1: Nearest rival rational for each just interval and the resulting jump in each formula. “Rival” is the simplest fraction within ± 50 cents that is distinct from the just ratio; error is the rival’s distance in cents.

Interval	Just	Rival	Err (¢)	ΔG	Δf	$\Delta \max$
Major second	9/8	64/57	2	+19	+72	+55
Minor third	6/5	77/64	2	+15	+73	+71
Major third	5/4	76/61	3	+74	+129	+71
Fourth	4/3	83/62	4	+109	+108	+79
Tritone	45/32	52/37	1	+37	+38	+7
Fifth	3/2	94/63	5	+54	+100	+91
Minor sixth	8/5	101/63	2	+103	+99	+93
Major sixth	5/3	103/62	3	+127	+127	+98
Minor seventh	9/5	16/9	1	0	+7	+7
Major seventh	15/8	118/63	1	+60	+110	+103

the ear pulling a mistuned pitch into the low-complexity well of its just target — the same well structure examined in Section 5 (*Comparison of the Three Formulae*).

Alternative equal temperaments

The tolerance model makes a natural prediction about which equal divisions of the octave (EDOs) are acoustically viable for which intervals. An n -EDO divides the octave into n equal steps of $1200/n$ cents; an interval is *well-approximated* when the nearest step falls within Δ_{tol} of the just target. The fifth (tolerance $\sim 10\text{¢}$ at A_3 , $\tau = 4$ Hz) is well-approximated by any EDO whose fifth is within 10 cents of 702 cents: this includes 12-TET (2¢ error), 19-TET (7¢), 31-TET (5¢), and 53-TET (0.07¢), but excludes 5-TET (720¢, error 18¢) and 7-TET (686¢, error 16¢). The major third (tolerance $\sim 6\text{¢}$ at A_3) is well-approximated only by 19-TET (1.5¢ error), 31-TET (0.8¢), and 53-TET (1.4¢); 12-TET (14¢) fails this test at most registers. This arithmetically explains the longstanding preference of renaissance-era theorists and contemporary microtonal composers for 19-TET and 31-TET: these systems preserve 5-limit harmony within the acoustic tolerance budget, whereas 12-TET can only do so for the fifth and fourth. The framework thus connects the arithmetic complexity measure $f(p/q)$ to a principled criterion for evaluating tuning systems.

The minor seventh and competing tunings

The minor seventh provides a striking special case. The Ptolemaic minor seventh 9/5 (996 cents) and the Pythagorean minor seventh 16/9 (995 cents) are only 1 cent apart, yet have distinct arithmetic characters. Gradus gives both the same value ($G = 9$), and is therefore unable to distinguish these historically competing tunings. The formula f does distinguish them: $f(9/5) = 13$ vs $f(16/9) = 20$, favouring the Ptolemaic tuning. The Pythagorean version $16/9 = (4/3)^2$ arises as two stacked fourths and is the minor seventh generated by the Pythagorean system, in which all intervals derive from the perfect fifth. Its higher f value reflects its larger numerator: the upper note sits at the 16th partial, far higher in the harmonic series than the 9th partial of the Ptolemaic version. Gradus, being symmetric in p and q , cannot see this distinction ($\Omega^*(16) + \Omega^*(9) = \Omega^*(9) + \Omega^*(16)$); the asymmetry of f is what exposes it.

9 A Partial-Beating Model of Perceptual Tolerance

The preceding section treats tolerance as a simple frequency deviation in cents. We now propose a qualitative model connecting partial-beating rates to the arithmetic structure of the interval. The components of this model — the connection between beating and roughness ([14]; [15]), the role of virtual pitch [12], and the perceptual salience of partials [11] — are individually well-established. The derivation of a $1/p$ tolerance scaling from the formula $f(p/q)$ is, to our knowledge, new, and should be understood as a qualitative conjecture pending experimental test.

The model

For a just interval p/q , the m -th coincident partial occurs at frequency $mp \cdot f_0$ (the (mp) -th harmonic of the lower note, with amplitude $\propto 1/(mp)$). If the actual interval deviates by Δ cents, the beat rate at the m -th coincidence is:

$$\beta_m \approx mp \cdot f_0 \cdot \frac{\ln 2}{1200} \cdot |\Delta| \quad \text{Hz}$$

The total roughness contribution, weighted by partial amplitude $\propto 1/(mp)$, is:

$$R \propto \sum_{m=1}^{\infty} \frac{1}{mp} \cdot \beta_m = \frac{f_0 \ln 2}{1200} \cdot |\Delta| \cdot \sum_{m=1}^M 1 = \frac{f_0 \ln 2}{1200} \cdot |\Delta| \cdot M$$

where the mp factors cancel. The sum diverges formally but is cut off in practice at the last audible partial $M \approx 10$ – 15 , making the total roughness roughly proportional to $p \cdot f_0 \cdot |\Delta|$ — the same scaling as the first partial alone. Focusing on the first coincidence (at frequency $p \cdot f_0$) is therefore a good approximation; the higher coincidences add a logarithmic correction that does not change the $1/p$ dependence.

Setting a threshold τ Hz on the first-coincidence beat rate gives a tolerance half-width:

$$\Delta_{\text{tol}}(p/q, f_0) \approx \frac{1730 \tau}{p \cdot f_0} \quad \text{cents}$$

The Plomp–Levelt critical bandwidth at frequency f is approximately $\text{CB}(f) \approx 1.72f^{0.65}$ Hz [15]; roughness is first perceptible at ~ 0.05 CB and reaches its maximum near 0.25 CB. At the first coincidence of the fifth ($p = 3$, frequency $3 \times 220 = 660$ Hz at A_3), $\text{CB}(660) \approx 118$ Hz, giving roughness onset at ~ 6 Hz. This calibrates the three thresholds used below: $\tau = 4$ Hz (below roughness onset, barely perceptible), $\tau = 8$ Hz (perceptible beating, not yet rough), and $\tau = 15$ Hz (at or above roughness onset for most registers).

Two properties follow. First, **tolerance is proportional to $1/p$** : the same numerator that appears in $f(p/q)$ determines how precisely the interval must be tuned to avoid audible beating. Consonant intervals (small p , small f) are not only perceived as smooth but are also forgiving of mistuning. Second, **tolerance depends on register**: at higher pitches the tolerance in cents narrows because the beating partials are at higher absolute frequencies.

Equal temperament in the model

Figure 2 shows the tolerance half-width for each standard interval at a reference register of $f_0 = 220$ Hz (A_3), for three threshold values $\tau \in \{4, 8, 15\}$ Hz. The 12-TET deviation from the just ratio is shown as a red horizontal mark.

The result is unambiguous for the fifth ($3/2$) and fourth ($4/3$): their ET deviations of 2 cents are well within tolerance at all three thresholds and at all musically relevant registers. The

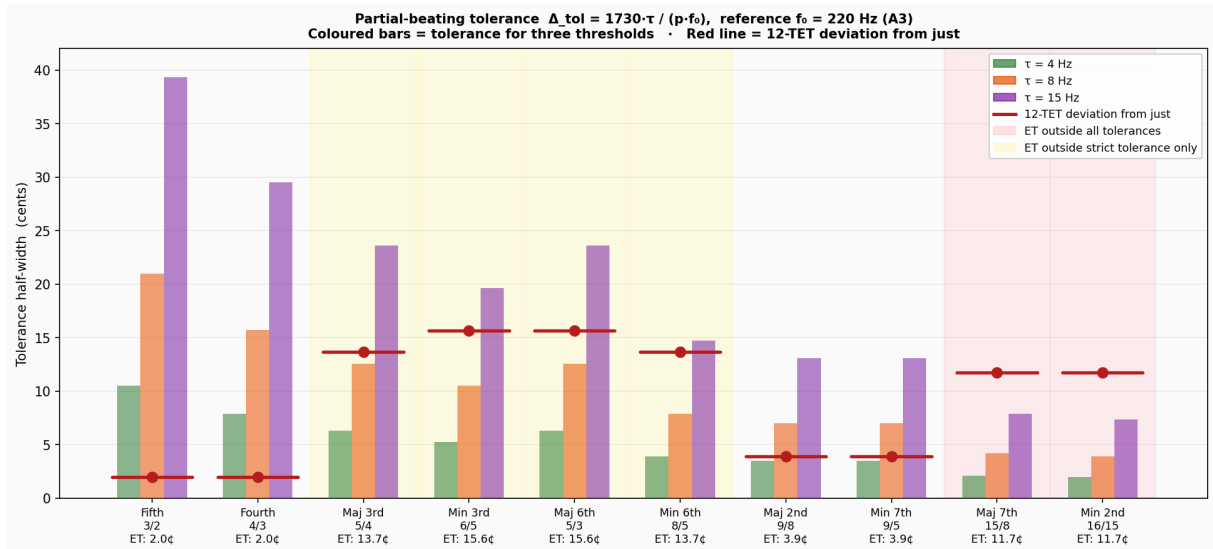


Figure 2: Partial-beating tolerance half-width vs 12-TET deviation for each just interval, at $f_0 = 220$ Hz (A3). Coloured bars show tolerance for three threshold values; the red mark is the 12-TET deviation. Yellow shading: ET exceeds the strict (4 Hz) but not the generous (15 Hz) threshold. Pink shading: ET exceeds all three thresholds.

thirds, sixths, and sevenths fall in a different category: their ET deviations of 14–16 cents substantially exceed the strict (4 Hz) threshold and sit near or above the generous (15 Hz) threshold. The major and minor seconds occupy an intermediate position. The major seventh (15/8) and minor second (16/15) are outside all three tolerance bands: the large numerator $p \in \{15, 16\}$ means even a modest deviation produces rapid beating.

Figure 3 plots the tolerance curves directly as functions of p , showing the $1/p$ hyperbolic structure. The interval points are coloured green (ET within all tolerances), orange (ET marginal), or red (ET outside all tolerances). The distribution confirms the pattern: the two green intervals — the fifth and fourth — are exactly those Western harmony has used as structural pillars throughout all periods of common practice, while the orange and red intervals are precisely those that show the greatest variation across tuning systems and historical periods.

Register dependence

Figure 4 shows how tolerance varies with register for four intervals, with vertical dotted lines marking the fundamental frequency at which the ET deviation crosses each threshold. For the fifth (top left), the crossing frequency exceeds 1000 Hz for the moderate threshold — in practice the ET fifth is acoustically tolerable throughout the entire playing range of most instruments. For the major third (top right), the 15 Hz crossing occurs at around 380 Hz (roughly G₄), meaning that in the upper half of the piano keyboard the ET major third produces beating comfortably above 15 Hz at the first coincident partial. For the minor third the situation is similar. The model therefore suggests that ET thirds may sound noticeably rougher in the upper register than the lower — a prediction consistent with the common practice of voicing thirds in the middle-to-lower register in orchestral writing and avoiding them in high woodwind chords.

The $1/p$ duality

The model suggests a duality between the two uses of p in this framework. In the dissonance formula $f(p/q) = p + \Omega^*(q)$, the term p is the *cost* of the interval: the effort required to

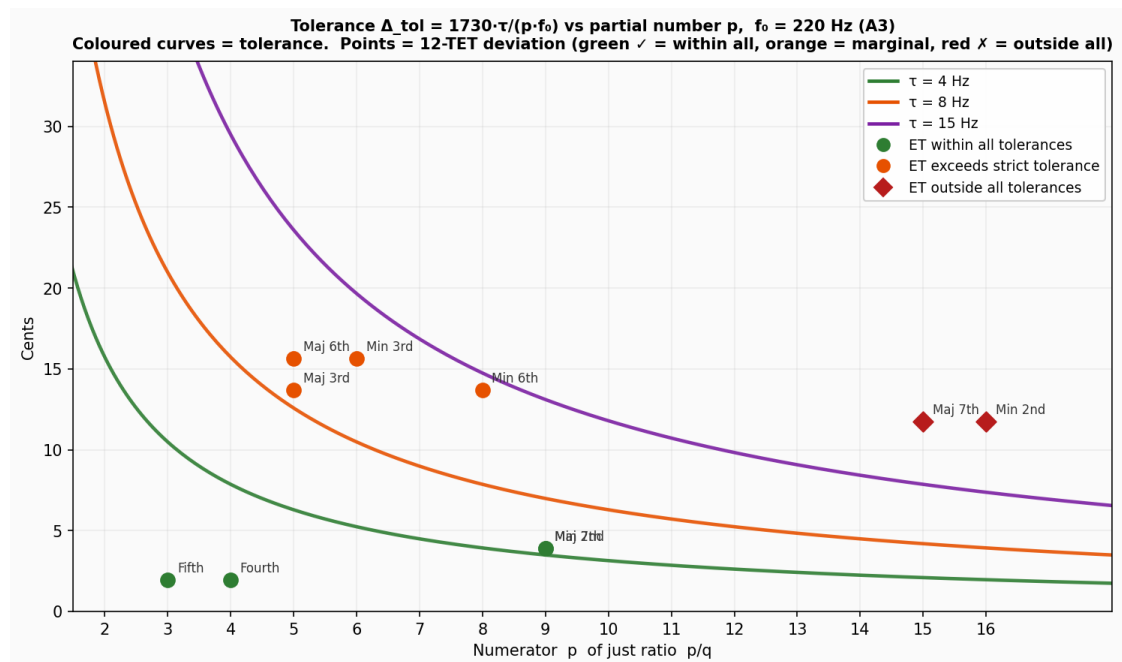


Figure 3: Tolerance half-width $\Delta_{tol} = 1730\tau / (p \cdot f_0)$ as a function of the partial number p at $f_0 = 220$ Hz. Coloured curves show three threshold values. Interval points show their 12-TET deviations: green = ET within all tolerances, orange = ET exceeds strict tolerance, red = ET outside all tolerances.

identify the upper note as the p -th partial of an implied fundamental. In the tolerance model, p appears in the denominator of Δ_{tol} : it is the *precision required* to keep the partial beating within an acceptable rate. These two roles of p are consistent with the same physical fact — that p is the partial number of the upper note in the harmonic series, and that both the cognitive effort of recognition and the acoustic sensitivity to mistuning may scale with partial height.

The implication for equal temperament is that Western musical culture has, over several centuries, adopted a tuning system in which the acoustically robust intervals (fifth and fourth, small p) are preserved almost exactly, while the acoustically fragile intervals (thirds and sixths, moderate p) are mistuned to the edge of — and in many registers slightly beyond — their acoustic tolerance. Whether this represents a deliberate optimisation, cultural habituation, or a combination of the two is a question that the arithmetic model alone cannot resolve.

10 Summary

Formula	ρ vs. human	Notes
Tenney height: $H = p \cdot q$	0.978	Symmetric; inverts 5/4 vs 5/3
Euler's Gradus: $1 + \Omega^*(p) + \Omega^*(q)$	0.979	Symmetric; two tie groups
Galileo / $\max(p, q)$	0.989	Simplest; one tie
$f(p/q) = p + \Omega^*(q)$	0.989	Asymmetric; two-stage conjecture
Optimised asymmetric weights	1.000	Six parameters; likely interpolation on 13 points
Roughness (Plomp–Levelt)	0.709–0.863	Register-dependent; fails on tritone*

* Helmholtz [14] introduced the beating-based dissonance curve that Plomp & Levelt [15] later parametrised from listener data; the two models produce equivalent rankings for our purposes.

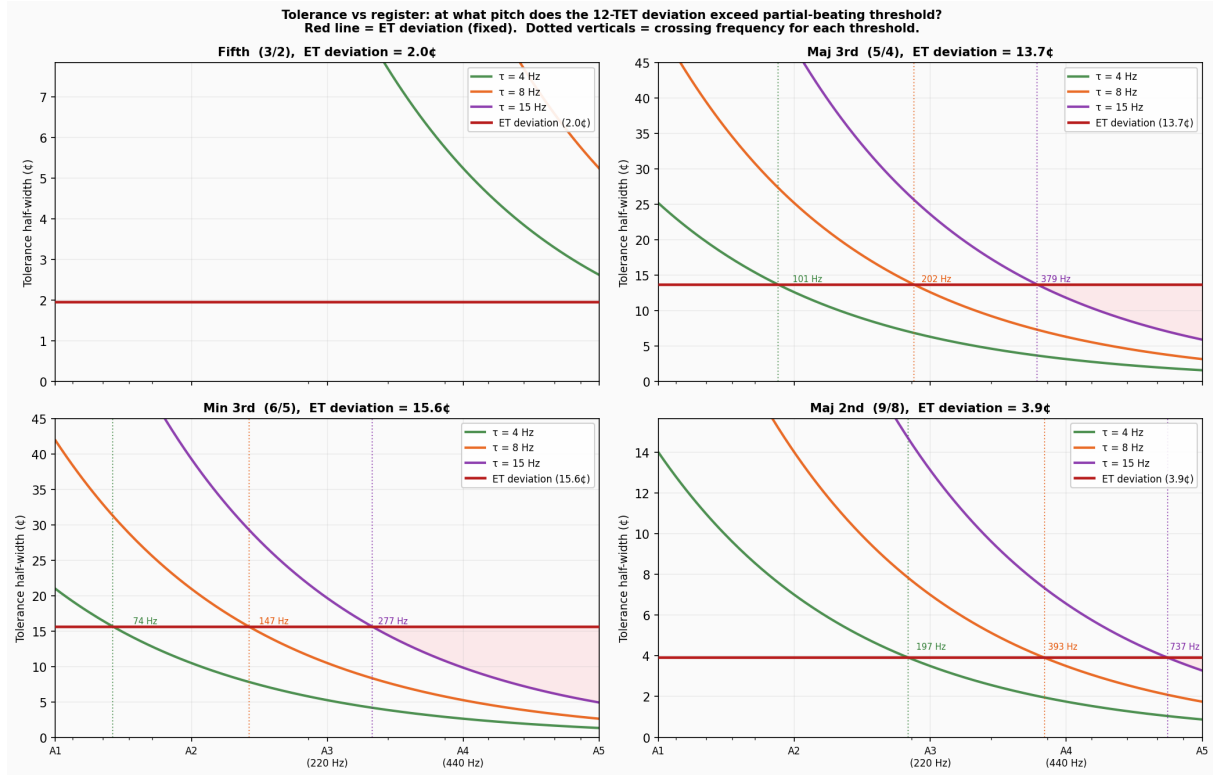


Figure 4: Tolerance half-width as a function of register (fundamental frequency, log scale) for four intervals. The red horizontal line is the fixed ET deviation; the coloured curves are the three threshold tolerances. Dotted verticals mark the register at which ET deviation crosses each threshold. Pink shading indicates the register range where ET exceeds the generous (15 Hz) threshold.

How close is f to the empirical optimum?

Allowing separate numerator and denominator weights for the primes 2, 3, 5 yields perfect rank agreement ($\rho = 1.000$) on this dataset, but with six free parameters fitted to 13 data points this likely reflects interpolation rather than structure. The zero-parameter formula $f(p/q) = p + \Omega^*(q)$ achieves almost the same correlation ($\rho = 0.989$).

The triangle $T(n, k)$

The formula $f(p/q) = p + \Omega^*(q)$ defines a triangular array of positive integers. Formally, $T(n, k) = n + \Omega^*(k)$ for $1 \leq k \leq n$ with $\gcd(n, k) = 1$, together with $T(1, 1) = 1$ for the unison. In OEIS notation this is *the triangle of $n + \Omega^*(k)$ read by rows, where for each row n the index k ranges over integers satisfying $1 \leq k \leq n$ and $\gcd(n, k) = 1$; only coprime pairs are listed, so rows have no blank entries and the row length equals $\varphi(n)$ for $n \geq 2$* . The triangle begins:

```

n = 1 : 1
n = 2 : 2
n = 3 : 3, 4
n = 4 : 4, 6
n = 5 : 5, 6, 7, 7
n = 6 : 6, 10
n = 7 : 7, 8, 9, 9, 11, 10
n = 8 : 8, 10, 12, 14
n = 9 : 9, 10, 11, 13, 15, 12

```

Here φ is Euler's totient function (OEIS A000010). The flat sequence of terms read left-to-right by rows begins:

1, 2, 3, 4, 4, 6, 5, 6, 7, 7, 6, 10, 7, 8, 9, 9, 11, 10, ...

The connection to Euler's Gradus (A275314) is: $T(n, k) = n + A275314(k) - 1$. The symmetric version of the triangle, $A275314(n) + A275314(k) - 1$, recovers Gradus.

The rightmost populated diagonal of the triangle, $T(n, n-1)$ for $n \geq 2$, gives the two-stage dissonance of the superparticular (epimoric) interval $n/(n-1)$ — the sequence of intervals between consecutive partials of the harmonic series: octave (2/1), fifth (3/2), fourth (4/3), major third (5/4), ... The diagonal sequence begins:

2, 4, 6, 7, 10, 10, 14, 12, 14, 16, 22, 17, 26, ...

Note that $T(8, 7) = 14 > T(9, 8) = 12$: the sequence is not monotone, because $\Omega^*(8) = \Omega^*(2^3) = 3$ is small despite 8 being large, making the major second 9/8 cheaper than the septimal interval 8/7. Both the triangle and this diagonal sequence are proposed as entries in the On-Line Encyclopedia of Integer Sequences (OEIS).

Several companion sequences arise naturally from the triangle and are noted here for potential future OEIS entries. The *row sums* $S(n) = \sum_{\substack{k=1 \\ \gcd(n,k)=1}}^n T(n, k) = n\varphi(n) + \sum_{\substack{k=1 \\ \gcd(n,k)=1}}^n \Omega^*(k)$ measure the total two-stage dissonance across all coprime intervals with denominator n ; the first sum $n\varphi(n)$ dominates, but the correction term reflects the prime structure of the coprime residues. The *row maxima* $\max_{k: \gcd(n,k)=1} T(n, k)$ and *row minima* $\min_{k: \gcd(n,k)=1} T(n, k)$ reveal the spread of dissonance values within each harmonic series level and are sensitive to the prime factorisation of n : rows where n is prime have $\varphi(n) = n-1$ entries all of the form $n + \Omega^*(k)$, while rows where n is a prime power have fewer coprime denominators with characteristically low $\Omega^*(k)$ values.

Table 2 shows all three formulae alongside the human consonance rank for the 13 standard just intervals of Western music, ordered by human rating. Ties in $f(p/q)$ and in $\max(p, q)$ are indicated by matching values; the sole unresolved tie (major third / major sixth, both $f = 7$) is discussed in Section 6.

A Code

All tables and figures in this paper were produced using Python; the complete code is available at <https://github.com/davidderoore/gradus>.

The following implementations define $\Omega^*(n)$, Euler's Gradus $G(p/q)$, the asymmetric formula $f(p/q) = p + \Omega^*(q)$, $\max(p, q)$, and the OEIS triangle $T(n, k)$ (Section 10). Each is self-contained and sufficient to verify any value in the paper or generate OEIS sequences.

PARI/GP

```
\\ Omega*(n) = sum of e*(p-1) over prime-power factors p^e || n
OmegaStar(n) = {
    if(n <= 1, return(0));
    my(f = factor(n));
    sum(i = 1, #f~, f[i,2] * (f[i,1] - 1))
};

\\ Euler's Gradus Suavitatis G(p/q)
Gradus(p, q) = {
```

Table 2: Dissonance values for the 13 standard just intervals, ordered by human consonance rank [1] (rank 1 = most consonant).

Interval	Ratio	$G(p/q)$	$f(p/q)$	$\max(p,q)$	$H(p/q)$	Human rank
Unison	1/1	1	1	1	1	1
Octave	2/1	2	2	2	2	2
Fifth	3/2	4	4	3	6	3
Fourth	4/3	5	6	4	12	4
Major third	5/4	7	7	5	20	5
Minor third	6/5	8	10	6	30	6
Major sixth	5/3	7	7	5	15	7
Minor sixth	8/5	8	12	8	40	8
Major second	9/8	8	12	9	72	9
Minor seventh	9/5	9	13	9	45	10
Major seventh	15/8	10	18	15	120	11
Minor second	16/15	11	22	16	240	12
Tritone	45/32	14	50	45	1440	13

$H(p/q) = p \cdot q$ is the Tenney height [9], the standard complexity measure in just intonation theory.
Spearman ρ against human rank: $G = 0.979$, $f = 0.989$, $\max = 0.989$, $H = 0.978$.

```

my(g = gcd(p,q)); p /= g; q /= g;
1 + OmegaStar(p) + OmegaStar(q)
};

\\ Asymmetric formula f(p/q) = p + Omega*(q)
f(p, q) = {
  my(g = gcd(p,q)); p /= g; q /= g;
  p + OmegaStar(q)
};

\\ Galileo / Farey metric max(p,q)
maxpq(p, q) = {
  my(g = gcd(p,q)); p /= g; q /= g;
  max(p, q)
};

\\ OEIS triangle T(n,k): row n, column k (1 <= k <= n, gcd(n,k)=1)
T(n, k) = if(gcd(n,k) == 1, f(n,k), 0);
tabl(nn) = for(n=1,nn, for(k=1,n, if(gcd(n,k)==1, print1(T(n,k),", "))))

```

Mathematica

```

(* Omega*(n) = Sum of e*(p-1) over prime-power factors p^e || n *)
OmegaStar[1] = 0;
OmegaStar[n_Integer /; n > 1] :=
  Total[#[[2]] (#[[1]] - 1) & /@ FactorInteger[n]];

(* Euler's Gradus Suavitatis G(p/q) *)
Gradus[p_, q_] := With[{r = p/q // Numerator, s = p/q // Denominator},
  1 + OmegaStar[r] + OmegaStar[s]];

```

```

(* Asymmetric formula  $f(p/q) = p + \Omega(q)$  *)
f[p_, q_] := With[{r = p/q // Numerator, s = p/q // Denominator},
  r + OmegaStar[s]];

(* Galileo / Farey metric  $\max(p, q)$  *)
MaxPQ[p_, q_] := With[{r = p/q // Numerator, s = p/q // Denominator},
  Max[r, s]];

(* OEIS triangle  $T(n, k)$ : first nn rows *)
T[n_, k_] /; GCD[n, k] == 1 := f[n, k];
Flatten @ Table[T[n, k], {n, 1, 9}, {k, 1, n}, GCD[n, k] == 1]

```

Python

```

from math import gcd
from sympy import factorint

def omega_star(n):
    """ $\Omega(n)$  = sum of  $e \cdot (p-1)$  over prime-power factors  $p^e \parallel n$ ."""
    if n <= 1: return 0
    return sum(e * (p - 1) for p, e in factorint(n).items())

def gradus(p, q):
    """Euler's Gradus Suavitatis  $G(p/q)$ ."""
    g = gcd(p, q); p, q = p // g, q // g
    return 1 + omega_star(p) + omega_star(q)

def f(p, q):
    """Asymmetric formula  $f(p/q) = p + \Omega(q)$ ."""
    g = gcd(p, q); p, q = p // g, q // g
    return p + omega_star(q)

def maxpq(p, q):
    """Galileo / Farey metric  $\max(p, q)$ ."""
    g = gcd(p, q); p, q = p // g, q // g
    return max(p, q)

# OEIS triangle  $T(n, k)$ : first nine rows
for n in range(1, 10):
    row = [f(n, k) for k in range(1, n + 1) if gcd(n, k) == 1]
    print(row)

```

B 12-TET Semitones and Just Intonation Targets

Table 3 shows each 12-TET semitone alongside its standard 5-limit just intonation target — the simple ratio that the tempered interval approximates in diatonic tonal music — and the dissonance formula values evaluated at that just target. The error column gives ET cents minus just cents (positive = ET is sharp of just). Formula values are evaluated at the just target, not the ET pitch.

Table 3: 12-TET semitones and their 5-limit just intonation targets. Error = ET cents – just cents (positive = ET is sharp of just). Formula values are evaluated at the just target, not the ET pitch.

Note	ET (¢)	Just target	Just (¢)	Error (¢)	G	f	H
C (unison)	0	1/1	0.0	0.0	1	1	1
C#	100	16/15	111.7	−11.7	11	22	240
D	200	9/8	203.9	−3.9	8	12	72
E♭	300	6/5	315.6	−15.6	8	10	30
E	400	5/4	386.3	+13.7	7	7	20
F	500	4/3	498.0	+2.0	5	6	12
F#	600	45/32	590.2	+9.8	14	50	1440
G	700	3/2	702.0	−2.0	4	4	6
A♭	800	8/5	813.7	−13.7	8	12	40
A	900	5/3	884.4	+15.6	7	7	15
B♭	1000	9/5	1017.6	−17.6	9	13	45
B	1100	15/8	1088.3	+11.7	10	18	120
C (octave)	1200	2/1	1200.0	0.0	2	2	2

The F# entry uses 45/32, the 5-limit diatonic tritone; as discussed in Section 5, the septimal tritone 7/5 ($f = 11$) would be a musically different just target with substantially lower complexity values.

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